The Pharmacist's Cipher II

A Morphological Framework, Visual-Textual Convergences,
and Testable Predictions for MS 408

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All hypotheses, interpretive decisions, and epistemic judgments are the author's.

Abstract

This companion report extends the statistical foundation established in Placa (2026)—hereafter "Paper I"—which presented six reproducible tests consistent with a pharmaceutical reading of the Voynich Manuscript (MS 408). Where Paper I deliberately restricted itself to transcription-only analyses, this report presents the full working model that those statistics support: a proposed agglutinative morphology organized around solvent-based prefixes and therapeutic class markers, together with visual-textual convergence analyses using Yale IIIF high-resolution images.

The report serves four purposes: (1) to describe the morphological framework in sufficient detail for independent evaluation, clearly separating structural facts from provisional glosses; (2) to document convergences between textual profiles and manuscript illustrations, including 17 working botanical identifications produced through a "sealed-envelope" protocol; (3) to present a record of 26 documented falsified hypotheses, demonstrating methodological discipline; and (4) to establish intellectual priority over 39 testable predictions at three confidence tiers, each with explicit falsification criteria.

This document is not a decipherment claim. It is a systematic inventory of the research directions that the framework opens, each recorded with its epistemic status, its anchorage to Paper I, and its falsification path. Its purpose is to fix priority, invite falsification, and provide a dated record of testable predictions. For each hypothesis, the report documents its consequences and falsification criteria, while reserving for future work the full exposition of specific convergence paths. Every section carries an explicit epistemic label ([STRUCTURAL], [FUNCTIONAL], [CONVERGENT], or [EXPLORATORY]) to ensure that no claim is presented at a higher confidence level than its evidence warrants.

Keywords: Voynich Manuscript, MS 408, agglutinative morphology, pharmaceutical codex, morphological framework, botanical identification, medieval pharmacopoeia, testable predictions

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How to Read This Report

This document is not a second proof paper. It is an atlas of disciplined hypotheses generated by a framework that Paper I has partially validated. Four principles govern its structure:

1. **This is not a decipherment claim.** The report proposes a working model, not a finished solution. Every section is explicitly marked for epistemic level.

2. **Claims are tiered by epistemic strength.** A five-level taxonomy is used throughout:

   - [STRUCTURAL] — measurable distributional fact, reproducible from the corpus
   - [FUNCTIONAL] — proposed role within the system, supported by distributional evidence and declared tests
   - [CONVERGENT] — interpretation supported by independent channels (text + image, or text + historical parallel)
   - [EXPLORATORY] — hypothesis with partial evidence and high potential yield
   - [FALSIFICATION CRITERION] — what would refute each claim

3. **Structural claims are sharply separated from glosses.** A segmentation claim (e.g., "*dy* is decomposable as *d+y*") is never presented at the same level as a semantic gloss (e.g., "*d* = medicine"). The former is distributional; the latter is interpretive.

4. **Many sections exist to establish priority and enable falsification,** not to assert conclusions. The report's value lies in making specific, testable, dated predictions—not in claiming they are already proven.

5. **This report is the second layer of a three-layer publication architecture.** Paper I establishes the statistical evidence that a coherent system exists. This companion documents each hypothesis the framework generates, together with its consequences and falsification criteria. Future dedicated studies will present the full convergence paths—visual, structural, codicological—that support each interpretation. This staged approach fixes intellectual priority while preserving the conditions for systematic verification.

Every section and subsection carries one of these status labels. They are not decorative.

Relationship to Paper I

Paper I (Placa, 2026) demonstrated six reproducible properties of the Voynich text consistent with a pharmaceutical register: cross-transcriber stability of morphological prefixes, a binary paradigm gap between *s-* and *sh-*, asymmetric lexical reuse between the Pharmaceutical and Stars sections, positional clustering of completion markers, section-specific morphological profiles, and a morpheme frequency hierarchy replicating pharmacological volume expectations. All six tests were computed from the transcription alone, with no dependence on illustrations or proposed glosses.

This companion report presents the morphological framework proposed to account for those patterns, together with visual-textual convergence analyses and a structured inventory of testable predictions. Nothing in this report duplicates Paper I. Where a Paper I result is referenced, it is cited by test number (e.g., "Paper I, Result 2") and not re-derived.

Part I
What Is Already Structurally Supported

This part presents the morphological framework's structural claims: segmentation rules, distributional facts, and positional constraints that are reproducible from the IVTFF corpus. Semantic glosses are presented separately from structural evidence.

1 Introduction and Scope

The Voynich Manuscript (MS 408, Yale Beinecke Rare Book and Manuscript Library) is a 15th-century codex written in an undeciphered script. Paper I established that its text contains internally consistent, non-random structure compatible with a medieval pharmaceutical register. This report asks: given that the text has grammatical, procedural, domain-compatible structure, what working model accounts for that structure?

The proposed account is an agglutinative morphology organized around solvent-based prefixes and therapeutic class markers, consistent with a medieval pharmaceutical register. It is offered as a working model, not as a proven decipherment. The framework makes specific, falsifiable predictions, and its value should be judged by whether those predictions survive independent testing—not by the elegance of its internal architecture.

The corpus is the IVTFF interlinear transcription (Zandbergen, 2022), primarily the Takahashi transcription (filter code `;H>`), containing 37,036 tokens across 5,206 lines. Tokens are separated by dots (`.`), not spaces. All counts use exact standalone matching (word-boundary), never substring matching. These conventions are identical to those in Paper I.

2 Morphological Architecture

The proposed agglutinative template is: **PREFIX + ROOT + SUFFIX**. Tokens are constructed by combining a limited set of prefixes (encoding solvent base or therapeutic class), roots (encoding core operations or substances), and suffixes (encoding processing parameters or states). This section presents the structural evidence for this template separately from the semantic glosses that have been tentatively assigned to its components.

2.1 Structural Segmentation Claims [structural]

These claims concern segmentability and distributional behavior. Each is verifiable from the corpus alone.

2.1.1 Prefix Families and Their Distributional Profiles

Four solvent-prefix families have been identified: *k*- (excluding *kh*-), *t*- (excluding *th*-), *l*-, and *s*- (excluding *sh*-). Note: the solvent morphemes are *k*, *t*, *l*, *s*; when preceded by the preposition *o* ("in"), they form the compound prefixes *ok*-, *ot*-, *ol*-, *os*-. Each family exhibits a distinct section-specific distributional profile. Paper I, Results 5 and 6 demonstrated that these profiles are systematically differentiated across the manuscript's six major sections and that the proportion of the morpheme *ee* across these families replicates a pharmacologically expected volume hierarchy (Spearman $\rho = 0.900$, $p = 0.042$ one-tailed).

Important asymmetry: The morpheme *o* co-occurs with the consonant it precedes at varying rates across the full consonant inventory: *k* 67.7%, *t* 70.7%, *r* 54.6%, *l* 53.6%, *p* 38.4%, *f* 40.2%, *d* 6.2%, *s* 4.5%, *y* 1.1%. This gradient shows that *o* is not exclusive to the solvent quartet but functions as a general preposition that combines with many consonants. The near-zero rate for *s*- is structurally significant: of ∼1,240 *s*- tokens (excluding *sh*-), only 85 occur

as *os-*, vs. 67–71% for *k/t/l*. Furthermore, the suffix profile after *s-* diverges sharply from that of *k-*/*t-* (overlap: 19.4%), with the discriminating suffixes *-eey*, *-edy*, *-eedy* at zero frequency after both *s-* and *l-*. The morphological split is thus *k-*/*t-* vs. all others—not "solvents vs. non-solvents."

This asymmetry is consistent with the Galenic pharmaceutical tradition, in which vinegar functions as a *reagent added to* preparations rather than as a *medium of dissolution*. The semantic frame "add vinegar to" does not take the locative preposition *o* ("in"), unlike "dissolve in water/oil/spirit." The bulk of the ∼1,240 *s-* tokens may reflect the broader Galenic category of "cold/cooling quality" (Arabic *bārid*), of which vinegar is the canonical vehicle. This is a single semantic axis at two levels of specificity, not two homonymous morphemes.

Four gallows identities (*p-*, *k-*, *t-*, *f-*) serve as first-token markers in the botanical folios, with distinct distributional profiles across sections.

2.1.2 Segmentation Claims

The following segmentation claims are supported by distributional evidence:

- *dy* decomposes as $d + y$: positional mean 0.715 within its line (late, consistent with a closure function), 69% of predecessors are action tokens, and *dy.dy* reduplications cluster on folios with toxic content.

- *ch* and *sh* are **unitary glyphs** (bench characters), not decomposable as *c+h* / *s+h*. The *h* is not a separable morpheme. Evidence: suffix overlap between *sh* and *ch* = 76.5%, whereas *sh* vs. *s* = 11.1%—the two bench characters form a class of their own. Paper I, Result 2 established a categorical paradigm gap (*sedy* = 0 across 37,036 tokens vs. *shedy* = 424). The two families encode complementary thermal values (cold/hot) with distinct section distributions.

- Simple vs. ornate gallows forms (e.g., *k* vs. *ckh*) represent a morphologically productive distinction, not calligraphic variation. The calligraphic model ("Model A") is falsified by three independent tests: (i) distributional chi-square across manuscript sections ($\chi^2 = 244.8$, $p < 0.001$); (ii) categorical exclusion of ornate forms from Position 1 in tokens (Δ 10–64 percentage points across gallows classes); (iii) suffix-profile divergence, with simple forms showing coefficient of variation 1.45–3.04 vs. near-zero for ornate forms. A further asymmetry is diagnostic: simple f/p tokens incorporate *ch* internally in 55–62% of cases (*f*: 62.0%, *p*: 54.8%), whereas ornate *cfh*/*cph* show internal *ch* in only 4–6% (*f*: 6.1%, *p*: 3.9%)—a 10–14× drop that cannot reflect calligraphic choice and implies structurally distinct token-internal composition. Notably, simple k/t tokens show ch-incorporation of only 25–30%, confirming that the elevated ch-incorporation rate is class-specific to f/p, not a general property of simple gallows.

- The suffix pair *-r*/*-l* shows directional precedence: *-r* consistently precedes *-l* in line order across all manuscript sections, with imperative tokens (*y-*) preferring *-r* (152 vs. 105).

- Coalescence rules: $qo + ok \rightarrow qok$, $qo + t \rightarrow qot$, $qo + l \rightarrow qol$. The compound forms show later positional means than their corresponding bare prefixes, consistent with verbal or processual function.

- The suffix *-am* is a morphologically productive maceration operator with positional mean 0.512—higher than any other tested suffix (*-dy*: 0.408; *-aiin*: 0.385; *-ol*: 0.357; *-or*: 0.333). Of 187 attested *-am* tokens, 58.3% appear after a solvent token in the same line, a rate comparable to the established closure operator *dy* (69.5%). Key productive forms: *otam* ($n = 16$), *dam* ($n = 28$), *cham* ($n = 8$), *okam* ($n = 7$), *ytam* ($n = 5$). The morpheme is almost absent from Balneo (< 0.1%), consistent with the observation that Balneo encodes the aqueous frame via the *ok-* prefix and does not require a separate immersion marker.

2.1.3 Thermal Asymmetry Across Gallows Classes [structural]

Corpus testing reveals a systematic thermal split between the four gallows classes that is independent of the simple/ornate distinction. Tokens anchored by *k* (aqueous) or *t* (oleic) show *ch*-adjacent rates of 66–69% (*k*: 68.7%, *t*: 66.2%), while tokens anchored by *f* (abortive/emmenagogue) or *p* (calming/sedative) show *sh*-adjacent rates of 58–59% (*f*: 58.4%, *p*: 59.0%), marginally exceeding their own *ch*-adjacent rates (55–56%). The chi-square for this k/t vs. f/p split on *ch*/*sh* collocational preference is significant at $p < 0.001$.

This is measured at three non-equivalent levels that must be kept distinct: (L1) *ch*/*sh* incorporated *inside* the gallows token itself; (L2) *ch*/*sh* in the *immediately following* token; (L3) *ch*/*sh* ambient anywhere in the line excluding the gallows token. The thermal asymmetry is robust at L1 and L2; L3 reflects section-level distribution more than local token semantics.

The bench characters *ch* and *sh* encode complementary thermal values (hot/cold) established in Paper I; the finding here is that the *gallows identity*—not just local phonotactics—conditions which thermal marker co-occurs. This is consistent with a reading in which the gallows prefix encodes the therapeutic class of the preparation (P-CALM, K-EXCIT, T-PROT, F-ABORT) and the thermal marker encodes a process parameter of that class. *Falsification criterion:* a permutation test showing that thermal-marker preference is independent of gallows class after controlling for section distribution.

2.1.4 Distributional Complementarity: Ornate *c_h* and Bench Characters *ch/sh* [structural]

A systematic distributional test reveals that the ornate gallows prefix (*c_h*, where the underscore denotes the base gallows letter) and the bench characters *ch*/*sh* in suffixes are in near-perfect complementary distribution. Ornate gallows tokens almost never carry *ch* or *sh* in their suffix, while simple gallows tokens frequently do:

Class	Ornate with *ch/sh* in suffix	Simple with *ch/sh* in suffix	Δ (pp)
k	1.5% ($N = 194$)	33.3% ($N = 1{,}159$)	−31.8
t	4.1% ($N = 489$)	41.2% ($N = 975$)	−37.1
p	4.7% ($N = 128$)	67.2% ($N = 545$)	−62.5
f	6.1% ($N = 33$)	72.6% ($N = 124$)	−66.5

Table 1: Distributional complementarity between the ornate prefix *c_h* and bench characters *ch*/*sh* in gallows suffixes. The near-total exclusion of bench characters from ornate suffixes (1.5–6.1%) against their high frequency in simple suffixes (33–73%) indicates that the ornate wrapping absorbs the thermal process information that would otherwise appear in the suffix.

This complementarity is paradigmatically confirmed: for 85–95% of ornate suffix types (representing 80% of ornate tokens), the corresponding simple form with *ch*+suffix or *sh*+suffix exists in the corpus. For example, ornate $ckh + y$ ($N = 37$) corresponds to simple $k + chy$ ($N = 29$) and $k + shy$ ($N = 5$); ornate $cth + ol$ ($N = 58$) corresponds to simple $t + chol$ ($N = 12$) and $t + shol$ ($N = 6$). The match rate covers 17/20 most frequent *k*-class and 19/20 most frequent *t*-class ornate suffix types.

Crucially, the paradigmatic counterparts skew toward *ch* (hot) over *sh* (cold): *k*-class counterparts show *ch*:*sh* = 3.00:1 and *t*-class counterparts show 3.73:1, both substantially above the corpus baseline of 1.84:1. Additionally, *sh*-initial tokens are specifically suppressed in the immediate context of ornate gallows (−5.2 pp after ornate vs. after simple), while *ch*-initial tokens show no such suppression. This asymmetry suggests that the ornate prefix encodes preferentially "hot" process information, though the transcription's binary categorization (ornate/simple) may collapse finer distinctions.

Interpretation. The ornate gallows wrapping *c...h* is not a calligraphic embellishment, a procedural classifier for instruments, or a cross-solvent transition marker. It is a **morphographic compression**: the thermal process marker (*ch* or *sh*) is absorbed into the glyph shape rather than spelled out in the suffix. This reduces redundancy while preserving information. The two encoding strategies—explicit ($k + chy$ = aqueous + hot-process) and compressed ($ckh + y$ = ornate-aqueous + process)—carry the same core information but occupy different syntactic positions: simple forms with *ch/sh* suffixes appear at Position 1 in 53.8% of cases (line launchers), while ornate forms appear at Position 1 in only 3.5% (medial/embedded operations).

Consequence for the writing system's semiotic architecture. The gallows character encodes three simultaneous channels of information: (1) *identity* (the base letter *p/k/t/f* = therapeutic class), (2) *thermal process* (the ornate wrapping = absorbed *ch/sh*), and (3) the *suffix* (specific processing parameters). This three-channel structure is consistent with the broader proposal that the manuscript's writing system is a compressed pharmaceutical notation, not an alphabetic cipher.

Falsification criterion: the complementarity collapses if ornate suffixes are shown to contain *ch/sh* at rates comparable to simple suffixes in a controlled corpus partition, or if the paradigmatic correspondence between ornate +X and simple *ch*+X / *sh*+X is shown to be an artifact of suffix frequency alone. *Open question:* the transcription collapses all ornate gallows into a single category per base class; the Petersen ascender chart shows 20+ ductus variants. A finer-grained paleographic catalog of ornate forms may reveal that distinct visual variants encode *ch* (hot) vs. *sh* (cold) separately, rather than collapsing both into one orthographic category.

2.1.5 The 224/224 Coverage Result

All 224 first tokens of botanical folios (100%) are decomposable via the agglutinative template without exception. The single apparent exception, f70r1, was a parsing artifact: the `@Cc` corpus tag was misread as content; the real first token of f70r1 is *okchedy* (line 2; line 1 is illegible), which decomposes as $ok + ch + edy$.

This coverage is presented as compatibility evidence, not as proof of the model. An agglutinative template with sufficient prefix and suffix options will inevitably accommodate a high proportion of tokens. The claim is that 100% coverage is achieved without ad-hoc rules or forced decompositions. *Falsification criterion:* any first token that resists decomposition without ad-hoc rules.

2.2 Provisional Glosses [functional]

The following are candidate semantic values proposed for the structural units identified in Section 2.1. They are informed by historical analogues in medieval Arabic pharmaceutical terminology and by domain-fit considerations. They are **not** entailed by the structural evidence and should not be confused with it.

"High-convergence" in the glosses below refers to internal model convergence—the degree to which a proposed meaning fits the observed distributional patterns—not to independently verified translations.

2.2.1 Solvent-Prefix Glosses

Prefix	Proposed gloss	Arabic analogue	Verbal form	% with *o-*	Tokens
k-	Aqueous base	mā’	*qok-*	67.7%	3,076
t-	Oleaginous base	duhn	*qot-*	70.7%	1,123
l-	Spirit base	rūḥ	*qol-*	53.6%	245
s-	Cold/acetous	khall	*qos-*	4.5%	∼15

Table 2: Solvent-prefix families with proposed Arabic correspondences and *o*-prefix rates. The near-symmetry of *k/t/l* (∼54–71%) vs. the anomalous *s* (4.5%) reflects the different semantic frame of the acetous/cold category: a reagent added to preparations, not a medium of dissolution. Token counts are standalone word-boundary matches in the full Takahashi corpus.

2.2.2 Therapeutic Class Glosses

The four gallows identities are proposed to encode therapeutic classes:

Gallows	Proposed class	Function
p-	P-CALM	Sedative / calmative
k-	K-EXCIT	Excitant / stimulant
t-	T-PROT	Protective
f-	F-ABORT	Abortive / emmenagogue

Table 3: Gallows identities with proposed therapeutic classes.

2.2.3 Morpheme Glosses

The following candidate values have distributional support but remain interpretive:

Morpheme	Proposed value and evidence
-am	Maceration/immersion operator: “[ingredient] submerged in [solvent]”. Positional mean 0.512—the highest of all tested suffixes (cf. *-dy*: 0.408, *-ol*: 0.357)—indicating consistent post-processual placement. 58.3% of *-am* tokens appear after a solvent token in the same line (vs. 17.6% with no solvent in the line). The dominant preceding solvent is oleic (*ot-*: 109 co-occurrences), followed by aqueous (*ok-*/*qok-*: 129 combined). Productive tokens: *otam* = macerate in oil; *okam* = macerate in water; *ytam* = immerse in oil (imperative *y-* + oleic *t-*); *sam* = macerate in vinegar (7 occ., Stars-dominant). Absence from Balneo (< 0.1%) is structurally expected: Balneo describes a patient in water, with the aqueous frame already encoded in the *ok-* prefix; no separate immersion marker is required. *Falsification criterion:* folio-level *-am* density should correlate with procedures involving soaking/steeping, not dry extraction.

Morpheme	Proposed value and evidence
o	General preposition "in" (Arabic *fī*): generates paradigm *ok* = *o*+*k* = "in water"; co-occurrence gradient across all consonants (k 67.7%, t 70.7%, r 54.6%, l 53.6%, p 38.4%, f 40.2%, d 6.2%, s 4.5%)
a	Surface preposition "on/upon" (Arabic $^{c}al\bar{a}$): Zodiac *a*/*o* ratio = 1.822 vs. Balneo *a*/*o* = 0.109
e/*ee*/*eee*	Three-grade quantity gradient, iconic (the number of *e*'s mirrors the quantity of substance). Grade 1 (*e*): "a little / q.s." (N_y = 1,986; N_{dy} = 2,801). Grade 2 (*ee*): "abundant" (N_y = 1,640; N_{dy} = 1,114). Grade 3 (*eee*): "copious" (N_y = 174; N_{dy} = 58). Grade 4 (*eeee*) is residual (4 tokens). All three productive grades are attested across both the -*y* and -*dy* series; each grade has lower token frequency than the previous one except Grade 1 in the -*dy* series, where -*edy* (N = 2,801) exceeds -*dy* (N = 2,117)—indicating that specifying at least "a little" is the default rather than the exception. The *ok*-/*ot*- prefix share doubles from 13–17% at grades 0–1 to 30–36% at grade 2, confirming that the quantity gradient preferentially targets the primary dissolution solvents (water, oil). Full paradigms are attested on common stems: e.g., *qok*-: ∅*y* = 130, *ey* = 101, *eey* = 299, *eeey* = 26. The volume hierarchy from Paper I, Result 6 is replicated within each grade. *Falsification criterion:* the gradient should produce no meaning contrast in a domain where quantity is irrelevant (e.g., astronomical notation). An earlier hypothesis of a universal intensity gradient was falsified (4/6 tests): the refined interpretation as prefix-specific quantity, strongest in dissolution contexts, supersedes it
e (in *eo*)	Airway / volatile / vapor: distinct from the quantity gradient above. Zodiac 39× enrichment, 78.7% of -*eo* lines contain zero water tokens. The relationship between *e*-as-quantity and *e*-in-*eo* is an open question: potentially the same morpheme ("subtle/fine amount" → "vapor/airway" when combined with *o* = "in"), or a homograph
eo	*e*+*o* = inhale / administer via airway: convergence with Cosmo visual analysis
d	Contextual pronominal referent (topic marker): antecedent shifts by section (plant in Herbal, patient in Balneo, formula in Stars). Distributional evidence: *d*- tokens = 14.93% of Herbal A, declining to 6.64% in Stars—a gradient consistent with the decreasing salience of a specific botanical referent. *daiin* (664 occ.) appears at line position 0 in 148 lines (2.84%), consistent with a topic-initial structure: "as for this [plant]—[instruction]". Generates: *dam* = referent macerated; *dal* = referent in alcohol; *dol* = referent in oil; *dar* = referent pure. *Falsification criterion:* single-plant folios should show higher *d*- density than composite or ambiguous folios.

Morpheme	Proposed value and evidence
pol	Calmant in alcoholic solution (probably laudanum): inter-sectional journey Herbal → Stars → Balneo, 100% line-initial in cross-references
r	Purity / essence marker (marks the exception; the unmarked default is compound/bound): co-occurs with solvents in 83.8% of lines—modifies the context, is not itself a solvent. Generates *eor* = vapors of the pure essence. Information-theoretic argument: in a pharmaceutical register, most preparations are compound; only "pure" needs marking
l	Spirit / alcohol, univocally throughout the system: solvent morpheme (Arabic *rūḥ*). Generates *eol* = *eo*+*l* = alcoholic vapors / distillation. An earlier positional disambiguation (prefix *l*- = spirit, suffix -*l* after *eo* = "bound") has been eliminated: *l* = spirit/alcohol in all positions
-*y*	Imperative / processual marker. Generates verbal forms across all prefix families. *Pragmatic note:* no separate morpheme for "mix" or "stir" exists in the system. In the pharmaceutical context, the default realization of "process!" is mixing/incorporation—the action of combining ingredients with solvents. The doubled form -*yy* is attested only 3 times corpus-wide (ratio to single -*y*: 4,901:1), twice on f86v4 (mnemonic meta-document): performative ritual closure, not a productive morpheme
raiin	*r*+*aiin* = "pure essence": *r* marks purity/exception (see above), *aiin* = base substance. Stars 4.1 ‰ dominant, *ch* co-occurrence 72.6%

Table 4: Morpheme glosses with distributional evidence. All values are candidate interpretations.

2.3 Two Orthogonal Pairs: *sh/ch* and *s/l* [structural]/[functional]

The morphological system operates on two orthogonal dimensions, each encoded by a distinct pair of morphemes:

Level	Pair	Meaning	Type
Process temperature	*sh* / *ch*	cold / hot	Unitary glyphs (bench characters, not decomposable)
Solvent / substance	*s* / *l*	acetous / alcoholic	Simple morphemes (single letters)

Table 5: Two orthogonal morphological pairs operating at different levels. The bench characters *sh* and *ch* encode the thermal axis of the process; the single-letter morphemes *s* and *l* encode the solvent/substance axis. Evidence for non-decomposability of *sh*/*ch*: suffix overlap *sh*↔*ch* = 76.5%, *sh*↔*s* = 11.1% (Paper I, Result 2).

This two-level architecture means that thermal information (how a substance is processed) and solvent information (what medium carries it) are encoded independently. A preparation can be cold-processed (*sh*) in an alcoholic medium (*l*), or hot-processed (*ch*) in an acetous medium (*s*), with no morphological conflict. The two pairs never collapse: *sh* is not a variant of *s*, and

ch is not a variant of *l*.

2.4 The Verbal Paradigms: *qok-*, *qot-*, *qol-* [structural]/[functional]

The compound prefixes *qok-*, *qot-*, and *qol-* are proposed to derive from coalescence of a verbal operator *qo-* with each solvent prefix. This claim has both a structural and a functional component.

Structural evidence: The three forms exist as productive prefix families with substantial corpus representation (3,076; 1,123; and 245 tokens respectively). Their positional means within lines are consistently later than those of the corresponding bare prefixes, a pattern consistent with verbal or processual function. A fourth form, *qos-*, is attested at approximately 15 tokens (standardized).

Functional interpretation: The proposed verbal status ("process with [solvent]") is a candidate reading supported by position and by the Stars section's dominance of *qok-*, compatible with a formulary register in which aqueous processing instructions predominate. This interpretation is not entailed by the positional data alone.

3 f86v4: Structural Anomaly and Operational Reading

3.1 Structural Anomaly [structural]

Folio f86v4 is a sub-folio containing a single rosette with four anthropomorphic figures at cardinal positions, surrounded by vegetation and depicted before a crescent moon. It is structurally anomalous in multiple respects:

- Density: 20.3 tokens per line, the corpus maximum.
- 189 tokens, 143 types, 41 hapax legomena (28.7% of types).
- Bipartite layout: lines 1–4 (`@Cc` annotation, dense) vs. lines 5–9 (`@P0` annotation, short).
- Line 1: 97.3% unique tokens, maximum entropy—structurally consistent with an index or categorical listing rather than narrative or recipe text.

Note: f86v4 is a single rosette with four figures. It is not f85r, which is a separate folio containing a 3×3 grid of nine rosettes. The two folios have been confused in prior work, including earlier stages of this research.

3.2 Operational Reading [convergent]

The four cardinal figures each perform a visually distinct action. Detailed visual analysis indicates that these actions are sequentially ordered and functionally non-redundant. The proposed reading is that the cycle North $\rightarrow$ East $\rightarrow$ South $\rightarrow$ West encodes a complete pharmaceutical production sequence—a closed operational cycle whose specific phases are identifiable from the figures' postures, gestures, and the objects they manipulate. The number and nature of these phases correspond to the functional categories attested in positional analysis (Paper I, Result 4: operations and vehicles precede materials, which precede completion markers).

The full figure-by-figure identification of each operational phase, together with its specific visual evidence from Yale IIIF high-resolution images, is documented and will be presented in a dedicated visual study. The present report fixes the hypothesis and its structural anchorage; the detailed convergence path is reserved for that future work.

Anchorage to Paper I: the positional ordering of morphological categories within recipe lines mirrors the proposed cardinal sequence. *Falsified if:* the four figures perform identical or unrelated actions under independent visual analysis, or if no sequential reading is sustainable.

3.3 Exploratory Interpretations and Implied Social Context [exploratory]

The following sub-hypotheses are progressively more speculative. They are registered here to establish intellectual priority and enable falsification, not as established conclusions. Each is summarized briefly; detailed evidence will be presented in dedicated future work.

Four exploratory readings extend the operational interpretation. **(a) Meta-document:** f86v4 may function as a categorical dictionary of the manuscript's morphological system—Line 1's 97.3% unique tokens and the bipartite layout (dense categorical lines followed by shorter operational lines) are consistent with an index, not recipe text. **(b) Mnemonic song:** the circular format, the anomalous hapax ratio (28.7%), the type/token ratio (0.757), and the terminal *yy* (only 3 tokens corpus-wide end in -*yy*; 2/3 are on f86v4) suggest a mnemonic device designed for oral transmission—sung, not read. **(c) Lunar/nocturnal context:** the four figures appear before a crescent moon, in vegetation, at night—compatible with the historically documented use of lunar phases as operational parameters for collection and fermentation. **(d) Restricted social milieu:** the encoding system (agglutinative cipher, non-representational illustrations, mnemonic oral transmission) implies knowledge that the producer considered worth protecting through deliberate concealment. The explicitly feminine iconography of the four operators is compatible with—but does not prove—a milieu in which pharmaceutical practice was associated with female practitioners or was transmitted outside institutional frameworks.

No biographical or identitarian claims are made. The social hypothesis is registered to fix priority over a consequential research direction; the full chain of visual and textual convergence supporting it will be presented in dedicated future work. *Falsified if:* the encoding system is better explained by purely commercial motivations (trade-secret protection within standard guild practice) with no residual social asymmetry.

Part II
What Appears to Converge Across Text and Image

This part presents interpretations that require both corpus analysis and visual analysis of Yale IIIF high-resolution images. Each section is marked for epistemic level. "Convergence" here means that independent channels (text and image) point in the same direction—it does not mean proof.

4 The Chromatic Code [convergent]

4.1 Proposed Therapeutic Color System

Analysis of the manuscript's botanical illustrations using Yale IIIF high-resolution images suggests a systematic chromatic code:

- **Blue** encodes the mental domain (psychotropic properties).
- **Black** encodes lethality (cross-class, not restricted to any single therapeutic category).
- **Yellow** (on roots) encodes lethality for underground plant parts, serving as an equivalent to black for subterranean structures.
- **Green** is the botanical default and carries no therapeutic information.

Evidence: multiple folios display chromatic patterns that are consistent with the pharmacological profiles derived from the textual analysis. In several cases, the color distribution on specific plant parts (roots, berries, leaves) aligns with the predicted therapeutic properties of the candidate plant. The strongest convergences involve folios where the textual profile independently predicts both psychotropic and lethal properties, and the illustration independently displays both blue and black on the same structure. Detailed folio-by-folio documentation of these convergences will be presented in a dedicated visual study using Yale IIIF high-resolution images.

4.2 Working Hypothesis: Deliberately Encoded Illustrations [functional]

The botanical illustrations are not realistic representations. They encode therapeutic *properties* (through color) while deliberately obscuring botanical *identity*. This is consistent with a coded pharmaceutical manual that allows an initiated reader to match preparations to plants, while preventing casual identification.

Plant identification within this framework proceeds in a fixed direction: textual processing profile → predicted pharmacological properties → predicted illustration colors → comparison with IIIF image. This directionality is methodologically critical: the illustration confirms the textual prediction, not the reverse. The "sealed-envelope" protocol described in Section 6 formalizes this approach.

5 A Proposed Administration Reading of the Cosmological Section (f85r) [convergent]/[exploratory]

5.1 The 3×3 Rosette Grid [convergent]

Folio f85r contains a grid of nine rosettes arranged in a 3×3 pattern. The grid structure is a visible fact; the functional mapping proposed below is an interpretation with partial textual support.

The four cardinal rosettes are proposed to encode four solvent domains:

- North = aqueous (supported: 29.4% *k*- tokens in textual content)
- South = oleaginous (partially supported)
- East = spirit (supported: f86v3 imperative prefix *y*- = 167–171 ‰, *ch/sh* opposition confirmed)
- West = acetous (open—no direct textual confirmation yet)

Two simultaneous axes are proposed: North–South encodes a dilution–concentration gradient (aqueous ↔ oleaginous), while West–East encodes a temperature gradient (cold/acetous ↔ hot/spiritous).

5.2 Angular Rosettes: Proposed Dual Function [exploratory]

The four angular rosettes (NW, NE, SE, SW) are proposed to serve a dual function:

As *transitions* between adjacent cardinal domains: NW mediates between aqueous and acetous, NE between aqueous and spiritous, SE between spiritous and oleaginous, SW between oleaginous and acetous.

As *administration ports* toward the center (patient/body): each angular rosette displays distinct visual features that are compatible with a specific route of administration attested in the medieval Arabic pharmaceutical tradition (liquid, vapor, insufflation, oral). The specific

visual-anatomical correspondences will be documented in detail in a dedicated study with full IIIF imagery.

A structural distinction is proposed: cardinal rosettes encode *frequency/dosage*, while angular rosettes encode *route of administration*.

5.3 Proposed Dosage Bundles (*Fasci*) [convergent]

Visual connections ("bundles" or *fasci*) between rosettes display systematic motifs:

- E–H bundles: chevron patterns 2+2 (confirmed via IIIF), proposed to encode topical application 2×/day.
- E–F and B–E bundles: tally marks 3×3, proposed to encode oral administration 3×/day.

Independent retrovalidation: the Balneo section's bipartition between topical and oral preparations yields a Spearman correlation of $\rho = -0.83$ with the predicted posological pattern. The retrovalidation is the strongest element of this claim; the specific dosage numbers (2×/day, 3×/day) remain interpretive.

5.4 Proposed Implicit Base Vectors [functional]

87.1% of Stars-section recipes contain no explicit solvent-prefix marker. The distribution of explicit markers is class-specific: K-EXCIT preparations show 0% explicit solvent markers (proposed default: spirit), while T-PROT preparations show 14% explicit markers (proposed interpretation: oil is non-default for this class and must be declared when used). The Cosmo section may provide class-specific default solvents, rather than a universal default.

6 Botanical Identifications [convergent]

6.1 Method: The Sealed-Envelope Protocol

A six-phase protocol was developed to discipline the identification process and prevent confirmation bias:

1. **Attribution hypothesis** — generated from the textual processing profile alone (prefix distribution, thermal markers, solvent ratios).
2. **Textual profile analysis** — detailed examination of solvents, therapeutic class, temperature indicators.
3. **Exclusion by incompatibility** — candidate plants eliminated if their known pharmacological properties contradict the textual profile.
4. **1-to-1 candidate confirmation** — a single candidate identified as the best match to the textual profile.
5. **Sealed envelope** — the predicted pharmacological profile generates predicted illustration colors (using the chromatic code), recorded *before* examining the illustration.
6. **Envelope opening** — comparison of predicted colors with the actual IIIF illustration.

Each identification receives a convergence score from 0 to 100. Scores reflect internal model convergence, not definitive botanical proof. Independent botanical and pharmacological verification is explicitly invited.

6.2 High-Convergence Working Identifications (Score $\geq$ 85)

Important methodological caveat: all scores below reflect internal model convergence—the degree to which the textual profile, the predicted pharmacological properties, and the illustration colors are mutually consistent within the proposed framework. They do **not** reflect independent botanical or pharmacological validation. The sealed-envelope protocol (Section 6.1) was designed and applied by the author; it disciplines the identification process but does not constitute a blind test. Independent replication by a specialist in medieval botany or pharmacology, using the textual profiles alone as input, is necessary to establish external validity.

Plant	Folio	Class	Score	Key convergence
Opium (*Papaver somniferum*)	f22r	P-CALM	90–95	—
Aconite (*Aconitum napellus*)	f10r	P-CALM	90–95	Yellow root = lethal
Nigella sativa	f16v	K-EXCIT	90–95	—
Cannabis bhang	f16r	P-CALM	90–95	Cold-water, *sh*-dominant
Cannabis hashish	f33v	T-PROT	90–95	Bipartite *ch/sh*, *dy.dy*
Belladonna (*Atropa belladonna*)	f17r	F-ABORT	85–90	BLU+NERO encoding
Peony (*Paeonia officinalis*)	f18v	T-PROT	85–90	—
Coffee (*Coffea arabica*)	f55v	K-EXCIT	85–90	—
Borage (*Borago officinalis*)	f50	P-CALM	85	—

Table 6: High-convergence working identifications. All scores reflect internal model convergence, not externally validated botanical proof. The top five candidates show convergence across text, image, and historical pharmacopoeia; the remaining four show strong but narrower convergence.

6.3 The Solanaceae Quartet [convergent]

A structural discovery of particular interest: four plants from the Solanaceae family (all containing tropane alkaloids) appear in four different therapeutic classes with four different preparation methods:

- f4r = Henbane (*Hyoscyamus niger*), K-EXCIT
- f17r = Belladonna (*Atropa belladonna*), F-ABORT
- f27v = Mandrake (*Mandragora officinalis*), F-ABORT
- f12 = Datura (*Datura stramonium*), candidate—removed folio

The key structural insight is that preparation temperature (*ch/sh* ratio) depends on the *therapeutic purpose*, not on the nature of the plant. The same thermolabile alkaloid receives opposite preparations (hot vs. cold) depending on the intended function. The manuscript classifies by *therapeutic function*, not by botanical family.

6.4 Lower-Confidence Working Identifications (Score 65–84) [exploratory]

Plant	Folio	Class	Score
Rosemary (*Rosmarinus officinalis*)	f21v	T-PROT	78
Henbane (*Hyoscyamus niger*)	f4r	K-EXCIT	75–80
Mandrake (*Mandragora officinalis*)	f27v	F-ABORT	75–80
Saffron (*Crocus sativus*)	f34v	K-EXCIT	75–80
Aloe vera	f3r	T-PROT	80–85
Wormwood (*Artemisia absinthium*)	f1v	K-EXCIT	70–75
Rose (*Rosa damascena*)	f48v	P-CALM	65–70
Cannabis tincture	f111r	K-EXCIT	OPEN

Table 7: Lower-confidence working identifications. Scores reflect internal model convergence.

6.5 The Bax Retrodiction Test [convergent] (with caveats)

This test is the closest approximation to partial external convergence currently available within the project. It has significant limitations that must be stated before the results: the sample is small ($n = 8$); the candidate space of medieval pharmacologically active plants is itself small, inflating chance-match probability; and the scoring rubric was designed by the author, not pre-registered or externally validated. These limitations mean that the quantitative result below should be read as *suggestive*, not as strong evidence.

With those caveats: eight blind predictions were generated from distributional profiles and compared with Bax's (2014) independently derived phonetic and visual identifications. Results: 3/8 exact species match (*Juniperus oxycedrus*, *Colchicum autumnale*, *Coriandrum sativum*), 2/8 family-level match, 1/8 profile-level match, 1/8 miss. Overall score: $24/40 = 60\%$; clean score (excluding the miss): $21/30 = 70\%$. A permutation test yields $p < 0.0005$, though this p-value is conditioned on the non-independent scoring rubric and should not be interpreted at face value.

To move this result from suggestive to evidential, two steps are necessary: (1) pre-registered scoring criteria, defined before the comparison; and (2) an external blind reviewer performing the scoring independently. Both are explicitly invited.

Part III
Predictions, Open Hypotheses, and Falsification Paths

This part is the core of the companion report. Its purpose is to: (1) establish intellectual priority over specific research directions; (2) demonstrate that the framework generates new, testable consequences—not only retroactive explanations; (3) invite verification and falsification; (4) provide a dated, attributable record of each prediction.

Tier	Level	Count	Testability
1A	STRUCTURAL	7	Corpus-only: reproducible in hours with Paper I scripts
1B	FUNCTIONAL	6	Corpus + framework interpretation
2	CONVERGENT	9	Corpus + Yale IIIF visual analysis
3	EXPLORATORY	17	Requires external expertise or new data
	Total	**39**	

Table 8: Summary of testable predictions by tier. Tier 1A predictions are independently verifiable from the IVTFF corpus; Tier 2 predictions require both corpus and visual analysis; Tier 3 predictions require external expertise (paleography, botany, Arabic pharmacology) or data not currently available to the author.

7 Tier 1A — Purely Structural/Distributional Predictions

These predictions are directly testable on the IVTFF corpus. An independent researcher can verify each one in hours with the scripts provided in Paper I's companion repository.

P1. [STRUCTURAL] The agglutinative template PREFIX + ROOT + SUFFIX decomposes 100% of first tokens across all 224 botanical folios. *Falsified if* any first token resists decomposition without ad-hoc rules.

P2. [STRUCTURAL] The four solvent-prefix families (*k*-, *t*-, *l*-, *s*-) exhibit complementary distributional profiles: each family dominates a different manuscript section. Note: *s*- shows a markedly different *o*-prefix rate (4.5% vs. 54–71% for *k/t/l*) and suffix profile (19.4% overlap with *k*-), consistent with a "cold/acetous" semantic axis rather than a pure solvent function. *Falsified if* any two families have identical section profiles.

P3. [STRUCTURAL] The verbal paradigms *qok-/qot-/qol-* derive from coalescence *qo + PREFIX* and show verbal positional behavior (later in lines than corresponding bare prefixes). *Falsified if qo*-forms appear earlier than bare forms.

P4. [STRUCTURAL] The morpheme *dy* is segmentable as $d + y$ with positional mean 0.715 and 69% action predecessors. *Falsified if dy* appears line-initially with $> 10\%$ frequency or if predecessors are predominantly nominal.

P5. [STRUCTURAL] The suffix pair *-r/-l* shows directional precedence: *-r* consistently precedes *-l* in line order across all sections. *Falsified if* any section shows *-l* preceding *-r*.

P6. [STRUCTURAL] Simple vs. ornate gallows forms are morphologically productive (not calligraphic variation). Model A is already falsified by three independent tests: distributional chi-square across sections ($\chi^2 = 244.8$, $p < 0.001$); categorical exclusion of ornate forms from token Position 1 (Δ 10–64 pp across gallows classes); suffix-profile divergence (CV 1.45–3.04 for simple forms vs. near-zero for ornate). Additionally, simple *f/p* incorporate *ch* internally in 55–62% of cases (*f*: 62.0%, *p*: 54.8%) vs. 4–6% for ornate *cfh/cph*, a structurally significant asymmetry. A further systematic split—*k/t* classes show 66–69% *ch*-adjacency (*k*: 68.7%, *t*: 66.2%) while *f/p* classes show 58–59% *sh*-adjacency (*f*: 58.4%, *p*: 59.0%)—indicates that gallows identity conditions thermal-marker co-occurrence independently of section distribution. *Strengthened by* the distributional complementarity analysis (Section 2): ornate gallows tokens carry *ch/sh* in their suffix at 1.5–6.1% vs. 33–73% for simple forms, with 85–95% paradigmatic correspondence between ornate +X and simple *ch*+X / *sh*+X—confirming that the ornate/simple distinction is a morphographic encoding of the thermal process marker. *Falsified if* ornate forms show identical distributions to simple forms in controlled environments.

P7. [STRUCTURAL] Cross-transcriber stability extends to all morphological claims in this report: every prefix and suffix falls within the 4% divergence margin established in Paper I, Result 1. *Falsified if* any claimed morpheme shows $> 10\%$ transcriber divergence.

8 Tier 1B — Structurally Grounded Functional Interpretations

These predictions involve distributional evidence interpreted through the proposed framework. They are structurally anchored but include a functional/semantic component.

P8. [FUNCTIONAL] The four gallows identities (*p/k/t/f*) map to four therapeutic classes with distinct section-specific distributions. *Falsified if* gallows distribution is uniform across sections.

P9. [FUNCTIONAL] The resultative *-am* appears in conservation/storage contexts (post-processual position). *Falsified if -am* tokens cluster in mid-process positions.

P10. [FUNCTIONAL] The preposition *o* generates the paradigm $ok = o{+}k$, with *o*-initial tokens showing locative distributional behavior distinct from processual tokens. *Falsified if o*-initial tokens show verbal/processual distribution.

P11. [FUNCTIONAL] The preposition *a* shows surface/destination distribution: Zodiac *a/o* ratio = 1.822 vs. Balneo *a/o* = 0.109. *Falsified if* the ratio inverts in independent corpus partitions.

P12. [FUNCTIONAL] Stars section functions as a semi-finished formulary that cross-references Pharma: 26:0 asymmetry (Paper I, Result 3) plus inter-sectional token journey Herbal → Stars → Balneo with line-initial constraint (100%). *Falsified if* Stars header tokens appear in Pharma.

P13. [FUNCTIONAL] *d* functions as a contextual pronominal referent whose antecedent shifts by section: *da-* successors differentiate sharply—Herbal → *chol/chor/cthor* (descriptive), Balneo → *shedy/ol/chedy* (active processes). *Falsified if da-* successor profiles are identical across sections.

9 Tier 2 — Predictions with Convergent Evidence

These predictions require both corpus analysis and visual analysis of Yale IIIF images. They are documented with sufficient detail for independent verification.

P14. [CONVERGENT] f86v4 encodes a meta-document: four figures at cardinal positions perform four sequential, functionally distinct operations that together constitute a complete pharmaceutical production cycle. The specific identity of each phase is documented and will be presented with full IIIF visual evidence in a dedicated study. *Falsified if* the four figures perform identical or unrelated actions under independent visual analysis.

P15. [CONVERGENT] The chromatic code encodes therapeutic properties: blue = mental domain, black = lethal, yellow (roots) = lethal underground equivalent. *Falsified if* blue-coded plants show no psychotropic properties in the historical materia medica, or black-coded plants are non-toxic.

P16. [CONVERGENT] The 3×3 rosette grid of f85r encodes an administration system with four cardinal solvent domains. North = aqueous (supported: 29.4% *k-*). *Falsified if* the textual content of cardinal rosettes shows no correlation with predicted solvent profiles.

P17. [CONVERGENT] The dosage bundles encode administration frequency: 2+2 chevrons = topical $2\times$/day, 3×3 tally marks = oral $3\times$/day. Retrovalidation: Balneo bipartition $\rho = -0.83$. *Falsified if* bundle patterns show no systematic variation between administration routes.

P18. [CONVERGENT] The sealed-envelope protocol identifies at least 9 plants with convergence score $\geq 85/100$. *Falsified if* an independent botanist/pharmacologist, given the textual profiles, consistently arrives at different plant candidates.

P19. [CONVERGENT] The Solanaceae quartet (f4r, f17r, f27v, f12) demonstrates functional classification over botanical classification: four tropane plants in four different therapeutic classes. *Falsified if* the four folios show identical morphological profiles.

P20. [CONVERGENT] The Bax retrodiction test (3/8 exact, $p < 0.0005$) represents partial convergence with an independent identification method. *Strengthened by* pre-registered scoring criteria and an external blind reviewer.

P21. [CONVERGENT] Cannabis appears in three distinct preparations across three sections: f16r bhang (P-CALM, cold water, *sh*-dominant), f33v hashish (T-PROT, bipartite *ch/sh*, *dy.dy*), f111r tincture (K-EXCIT, *l*- = 94.3 ‰). *Falsified if* the three folios show identical morphological profiles.

P22. [CONVERGENT] The morpheme *e* marks airway/volatile contexts, with *eo* = *e*+*o* = "administer via airway." Zodiac 39× enrichment, 78.7% of -*eo* lines without water tokens, Pharma depleted. *Note:* the quantity gradient *e/ee/eee* (see morpheme table) uses the same grapheme; the relationship between *e*-as-quantity and *e*-in-*eo* is an open question—potentially the same morpheme ("subtle amount" → "vapor" when combined with *o* = "in"), or a homograph. *Falsified if* *e*-containing tokens show no distributional difference from non-*e* tokens in volatility contexts.

10 Tier 3 — Exploratory Hypotheses

These are the furthest-reaching predictions of the framework. They are documented here to establish intellectual priority and to provide specific research directions. Each carries lower confidence but high potential yield. They should be read as motivated conjectures, not as claims.

P23. [EXPLORATORY] *(high yield)* H8b: the ductus (stroke path) of each gallows traces the physical path of processed material across the 3×3 grid of f85r. At least one gallows variant has been traced through its rosette sequence using the Petersen ascender chart, yielding a multi-station path consistent with a specific pharmaceutical transformation. Proposed three-layer semiotic structure: illustrations = ingredients, gallows = process, text = parameters. Supporting distributional evidence (not dependent on the rosette topology claim): (i) gallows class conditions thermal-marker co-occurrence—k/t classes prefer *ch* (66–69%) while f/p classes prefer *sh* (58–59%)—consistent with the model's prediction that class identity encodes a process axis; (ii) the simple/ornate distinction encodes a morphographic compression of the thermal process marker (see Section 2, "Distributional Complementarity"), confirming that the glyph shape carries process information beyond the base letter identity; (iii) the three-channel encoding model (identity + thermal process + parameters) is independently confirmed by the complementarity data without any reference to the rosette topology. The topology and rosette-path claims require a dedicated *paleographic ductus catalog*: a systematic encoding of each Petersen variant's stroke path as a directional sequence on the 3×3 grid, enabling computational comparison across the 20+ known ductus variants. This catalog is the necessary precondition for any corpus-level test of the topological claim and does not yet exist. *Falsified if* gallows variants show no spatial correlation with rosette positions, or the ductus catalog reveals no systematic correspondence between stroke direction and rosette adjacency. *Note: this remains one of the most speculative predictions in the report. It is included because the three-channel encoding model (now confirmed distributionally) makes the topological extension a natural—and testable—next step.*

P24. [EXPLORATORY] *(partially superseded)* The original hypothesis that ornate gallows encode specific instruments or apparatus has been *superseded* by the distributional complementarity finding (Section 2, "Distributional Complementarity"): the ornate wrapping *c*...*h* functions as morphographic compression of the thermal process marker *ch*/*sh*, not as an instrument classifier. However, the distributional *variation among* ornate forms remains real: the -*eo* gradient across

ornate classes shows Fisher $p < 0.01$, and each ornate class occupies a distinct suffix profile. The residual variation, after accounting for the absorbed thermal marker, may encode finer processing distinctions that remain to be characterized—but the instrument interpretation is no longer the preferred reading. *Falsified if* the residual distributional differences among ornate classes are shown to be section artifacts.

P25. [EXPLORATORY] The missing folios (14 across 6 gaps) were deliberately removed, not lost to wear. Adjacent folios carry toxic markers: f11 *dy.dy+pol*, f90 *poleeol+sam*, f108 maximal density 1,061 tokens. f12 is a candidate for *Datura stramonium*. *Falsified if* codicological analysis shows the gaps are consistent with mechanical damage rather than excision.

P26. [EXPLORATORY] f13r constitutes a post-exposure contrast protocol compatible with Ibn Sīnā's anti-tropane treatment: T-PROT, *ch/sh* = 0.43 (strongly cold), *s-* = 3 (cold/acetous), *l-* = 0 (zero spirit), *y-* = 7 (imperatives). The bigram *s.y* appears 2/3 corpus-wide in f13r alone. *Falsified if* the morphological profile is compatible with a standard recipe rather than an emergency protocol.

P27. [EXPLORATORY] The *lo/la* distinction encodes administration route: *lo* = oral, *la* = topical. Balneo spirit *lo/la* = 10.2 (oral dominant), water *oko/oka* = 0.11 (topical dominant). *Xa* position always > *Xo* position (4/4 solvents, $p = 6.25\%$). *Falsified if* the positional and ratio patterns are consistent with random morphological variation.

P28. [EXPLORATORY] The *eo*-system encodes four post-processing states: *eol* = *eo* + *l*(spirit) = alcoholic vapors / distillation, *eor* = *eo* + *r*(pure) = vapors of the pure essence, *eos* = acetous vapor form, *eody* = vaporize (the only verbal member). Note: *l* = spirit/alcohol univocally throughout the system (the earlier positional disambiguation of *l* is eliminated); *r* = pure/essence, marking the exception (the unmarked default is compound/bound). 82–93% bare terminal for nominal members, 68.6% *+y* verbal for *eod*. ∼1,700 tokens recovered. *Falsified if* the four members show identical distributional behavior.

P29. [EXPLORATORY] Implicit base vectors are class-specific: K-EXCIT default = spirit (0% explicit), T-PROT default = oil (14% explicit). The Cosmo may provide these defaults. *Falsified if* all classes show identical rates of explicit solvent declaration.

P30. [EXPLORATORY] The Zodiac section encodes melothesia (astrological-medical correspondences), not a pregnancy calendar or astronomical observations. Zodiac *ot-* enrichment = 2.43× baseline, near-zero *qok-*/*qot-*. *Falsified if* the content is better explained by astronomical rather than medical-astrological models.

P31. [EXPLORATORY] *pol* encodes a calmant in alcoholic solution (probably laudanum) and traces an inter-sectional journey: Herbal (definition) → Stars (standardized preparation) → Balneo (therapeutic application). 100% line-initial in cross-reference contexts. *Falsified if pol* tokens show no inter-sectional distributional pattern.

P32. [EXPLORATORY] The Cosmo two-axis model: N–S = dilution ↔ concentration, W–E = cold ↔ hot. Angular rosettes represent transitions between adjacent cardinals. *Falsified if* the textual content of angular rosettes shows no transitional properties.

P33. [EXPLORATORY] The angular rosettes of f85r serve as administration ports, each encoding a distinct route of administration attested in the medieval Arabic pharmacopoeia. Four routes are proposed, corresponding to the four angular positions; each is visually distinguishable in the IIIF images. *Falsified if* independent visual analysis identifies no anatomically or functionally distinct features across the four angular rosettes.

P34. [EXPLORATORY] The circular text of f86v4 constitutes a mnemonic covering the manuscript's lexicon, implying oral transmission. Terminal double *-y* + closing formula. Corpus-wide, only 3 tokens end in *-yy* (ratio to single *-y*: 4,901:1); 2/3 appear on f86v4 (*shyy* line 3, *yy* final token).

The bare *yy* functions as a ritual seal—a doubled imperative closing the song. *Pragmatic note:* the imperative *-y* ("process!") has no distinct "mix/stir" morpheme in the system; in recipe context, the unmarked realization of "process!" *is* mixing/incorporation, since combining ingredients with solvents is the default pharmaceutical action. The doubled *yy* at the song's close thus carries the pragmatic force of "mix! mix!"—a compressed performative closing. *Falsified if* the hapax ratio (28.7%) and type/token ratio (143/189) are consistent with normal recipe text.

P35. [EXPLORATORY] f85r rosette 2 (inner ring) serves as a Stars index: 22 petals $\approx$ 23 Stars folios. *Falsified if* the petal count does not correspond to the folio count under any reasonable interpretation.

P36. [EXPLORATORY] Eight pinnate-leaved plants in the Herbal form a structurally obligatory cohort with shared morphological features. *Falsified if* the pinnate cohort shows heterogeneous morphological profiles.

P37. [EXPLORATORY] The paleographic backlog (H1–H6B) contains confirmable predictions: H4 (stars with tail = modifier) requires IIIF verification; H6B (q = prefixal operator) contributes to the *qo*-coalescence model. Each has specific visual or distributional tests.

P38. [EXPLORATORY] *(methodological prerequisite)* A paleographic ductus catalog of all ornate gallows variants—encoding each Petersen ascender-chart variant as a directed stroke sequence on the 3×3 grid—will reveal that visually distinct ornate sub-types map to distinct thermal processes. Specifically: some ornate variants should correspond to *ch*-type (hot) processes and others to *sh*-type (cold) processes, resolving the open question noted in the complementarity analysis (Section 2). The current transcription collapses all ornate forms into a single category per base class, masking this predicted sub-distinction. The catalog requires high-resolution IIIF images, paleographic expertise, and a formal encoding scheme that captures stroke direction, loop orientation, and the position and shape of the alpha ornament. This prediction is registered to establish priority and to define the minimal requirements for computational validation of the ductus-encoding hypothesis (P23). *Falsified if* the catalog reveals no systematic correspondence between ornate sub-types and thermal context (*ch* vs. *sh* in adjacent tokens).

P39. [EXPLORATORY] The terminal *yy* of f86v4 is a ritual closure marker of the mnemonic song—a doubled imperative with performative (not morphological) function. Only 3 tokens ending in *-yy* exist in the entire corpus (37,036 tokens); 2/3 are on f86v4 (*shyy* on line 3, bare *yy* as the final token). The 4,901:1 ratio of single *-y* to doubled *-yy* excludes productive morphological status and points to an intentional pragmatic device. The pragmatic reading "mix! mix!" derives from the observation that the imperative *-y* ("process!") has no competing "mix" morpheme in the system: in a pharmaceutical context where combining ingredients with solvents is the default action, "process!" *is* "mix!". *Falsified if* additional *-yy* tokens are found to be productive in recipe sections (indicating morphological rather than performative function), or *-yy* is shown to be a transcription artifact rather than a distinct glyph sequence.

Part IV
Falsification Record and Epistemology

11 Twenty-Six Falsified Hypotheses

This section exists for two reasons: to demonstrate methodological discipline—rare in Voynich studies, where hypotheses are typically proposed and then abandoned without record—and to prevent future researchers from re-exploring dead ends.

#	Hypothesis	Falsification
1	*-am* = *marham* ("ointment")	Falsified (session 3c). Replaced by *-am* = resultative participle
2	Zodiac = pregnancy calendar	Falsified. Replaced by melothesia
3	Stars = astral section	Falsified. Replaced by semi-finished formulary
4	Blue = psychotropic strict	Falsified. Broadened to mental domain
5	Glyphs = complete Ghubār numeral system	Falsified (session 57): 7/10 matches trivial
6	Author profile Maghreb/al-Andalus on glyph basis	Falsified (session 57)
7	*ee/eee* = universal intensity gradient	Falsified on 4/6 tests. Refined to three-grade prefix-specific quantity (*e* = a little, *ee* = abundant, *eee* = copious); *ok-/ot-* share doubles at grade 2. See morpheme table
8	*eo* = evaporate/reduce	Falsified (session 46): Zodiac 39× enriched. Replaced by *eo* = inhale
9	Simple/ornate gallows = calligraphic (Model A)	Falsified: $\chi^2 = 244.8$ ($p < 0.001$); positional Δ 10–64 pp; suffix CV 1.45–3.04; *ch*-incorporation 55–62% (simple *f/p*) vs. 4–6% (ornate)
10	Black = abortive cross-class	Falsified (session 47). Replaced by lethal cross-class
11	*kol* = squill	Falsified
12	*ot-* = body	Falsified
13	*dy* = transitive "make medicine for [target]"	Falsified (session 8). Replaced by intransitive *d+y*
14	Illustrations = placeholders	Falsified
15	f50v = mandrake	Falsified. Reassigned to f27v
16	f14v = rose	Falsified. Reassigned to f48v
17	f54r = henbane	Falsified. Reassigned to f4r
18	*sam* = toxic concentration	Falsified
19	F-ABORT = terminal class	Falsified
20	f55r = rose	Falsified (IIIF: white flowers, blue pistils)
21	*okeody* = hellebore (Bax plant name)	Falsified: procedural token *ok+eo+dy*, never folio header
22	Green leaves = calmative	Falsified: green = botanical default
23	*-ed* = separate participle from *-edy*	Falsified: identical distribution
24	*-el/-er* paradigm	Falsified (session 46): *-el* = 8 tokens, no productivity
25	f17r = rue/artemisia	Superseded by belladonna (session 62)

#	Hypothesis	Falsification
26	Ornate gallows = cross-solvent transition markers	Falsified (session 67): 7 independent distributional tests showed no signal or negative signal for cross-solvent encoding. Controlled for section: ornate lines show equal or lower solvent diversity. Replaced by morphographic compression of thermal marker

Table 9: Complete falsification register. "Replaced by" indicates the current hypothesis that superseded the falsified one. "Superseded" indicates the old identification was displaced by a better candidate.

12 Epistemological Framework

This section is placed after the evidence, but its principles governed the entire research process. It should be read as a declaration of method, not as post-hoc justification.

12.1 Core Principles

The research followed a strict epistemic protocol: data → hypothesis → test, never the reverse. Every hypothesis was required to have a declared falsification criterion before being tested. When a hypothesis was falsified, it was recorded explicitly (Section 12) and all downstream claims that depended on it were updated or retracted.

Three levels of claim are distinguished throughout: **structural facts** (measurable distributional properties reproducible from the corpus), **functional roles** (proposed roles within the system, supported by distributional evidence), and **semantic glosses** (proposed translations or meanings). These levels carry different confidence and must never be conflated.

The confidence scale is asymmetric: it adjusts downward after falsification, never upward without a new positive test. This asymmetry is deliberate and is meant to counteract the natural tendency toward confirmation.

12.2 The Gap Between Internal and External Strength

The framework's current assessment is 95/100 for internal coherence and 63/100 for external probative force. The gap is significant and is not minimized here.

Internal coherence means that the model's components are mutually consistent: the morphological template accounts for the distributional patterns, the prefix families show complementary profiles, the positional constraints follow from the proposed functional categories. This is necessary but not sufficient.

External probative force means the ability to convince an independent researcher who does not share the model's assumptions. The gap exists because the morphological system is *domain-compatible*, not *domain-specific*: an agglutinative model with four prefixes and four initial classes is compatible with any domain that involves categories, operations, and parameters.

Why pharmaceutical and not another domain? The morphological template alone does not select for pharmacy over alchemy, cuisine, textile dyeing, or any other craft organized around materials, processes, and parameters. What constrains the interpretation to the pharmaceutical domain is the *convergence of five independent channels*, none of which is individually sufficient:

1. **Recipe-like text structure.** Paper I demonstrated that the text exhibits procedural sequencing, section-specific enrichments, and positional ordering consistent with recipe compilations—not narrative, liturgical, or astronomical prose (Paper I, Results 3–6).

2. **Illustration convergence.** The sealed-envelope protocol (Section 6) generates pharmacological predictions from textual profiles that independently match chromatic features in the IIIF illustrations. A culinary or textile reading would not predict blue = psychotropic or black = lethal on specific plant parts.

3. **Bax phonetic convergence.** Three of eight blind retrodictions match Bax's independently derived plant identifications at the species level (Section 7). The matched candidates are pharmacologically active plants (juniper, colchicum, coriander), not food crops or dye plants.

4. **Sectional organization.** The manuscript's division into Herbal, Stars, Balneo, and Cosmo sections maps coherently onto the stages of pharmaceutical production (ingredient identification, formulation, therapeutic application, administration system)—a mapping that has no obvious analogue in other craft traditions.

5. **Historical fit.** The solvent quartet (water, oil, spirit, vinegar) and the therapeutic class system (calmative, excitant, protective, abortive) correspond to attested categories in the medieval Arabic pharmaceutical tradition, particularly the *aqrābādhīn* genre (see al-Kindī, al-Rāzī, Ibn Sīnā).

No single channel proves the pharmaceutical reading. But the five channels are independent: the text structure does not determine the illustration colors; the illustration colors do not determine the Bax matches; the Bax matches do not determine the sectional organization. The convergence of independent channels is the framework's principal argument against the domain-compatibility objection. It remains, however, an argument from convergence—not a proof. The "killer test" (Section 14) defines what would constitute proof.

12.3 On Working with AI

Computational analysis was performed with Claude (Anthropic), an AI language model. The collaboration is disclosed fully and without reservation.

The AI's role was: corpus processing, statistical computation, systematic hypothesis testing, and documentation. The human researcher's role was: research intuitions, interpretive leaps, direction of inquiry, and final epistemic judgment on every claim.

This collaboration represents AI as research accelerator, not as oracle. The model does not "believe" its outputs; the human researcher takes responsibility for every claim in this report, including errors. The 26 documented falsified hypotheses in Section 12 attest that the process was not one of passive acceptance of AI-generated suggestions but of active, adversarial testing.

13 What Would Definitively Confirm or Refute This Framework

External confirmation paths:

- Independent paleographic verification of the agglutinative segmentation by a specialist in medieval manuscript scripts.
- Identification of a known Arabic pharmaceutical text (from the *aqrābādhīn* genre) with structural parallels to the distributional patterns documented here.
- Botanical expert verification of at least three of the nine high-convergence plant identifications through an independent sealed-envelope protocol.

- Cross-linguistic comparison: processing the text of a known medieval pharmaceutical manuscript through the same morphological template to test whether analogous distributional signatures emerge.

The "killer test": A known medieval pharmaceutical text, processed through the morphological template described in Section 2, should show analogous distributional signatures: section-specific prefix enrichment, paradigm gaps, and positional ordering of functional categories. If it does not, the template's pharmaceutical specificity is weakened. The author considers this the single most important next step and actively invites specialists in medieval Arabic pharmaceutical manuscripts to attempt it.

What would falsify the entire framework:

- A demonstrated non-pharmaceutical text that produces identical distributional patterns under the same analysis pipeline.
- A successful, independently verified decipherment of the Voynich Manuscript yielding non-pharmaceutical content.
- Proof that the agglutinative segmentation is arbitrary: that random prefix assignment produces equally good coverage rates and paradigm gaps.

The author welcomes all three types of falsification attempt. The corpus, the transcription, and the analysis scripts for Paper I are publicly available. Any researcher who wishes to challenge, extend, or independently replicate the framework has the materials to do so.

14 Cautious Notes on the Implied Producer

This section is deliberately brief and limited to functional inferences. It does not make biographical, ethnic, or identitarian claims about the manuscript's author(s). The risk of such claims is that they become the most citable—and most attackable—element of the report, distracting from its scientific merits.

The framework's structural properties, taken together, imply a producer profile defined by *competence*, not identity:

- **Practical pharmaceutical competence.** The morphological system encodes solvent-specific operations, thermal processing distinctions, dosage markers, and therapeutic class assignments that are consistent with working knowledge of pharmaceutical production, not with theoretical learning alone.
- **Compilatory orientation.** The document organizes existing knowledge—it classifies, cross-references, standardizes. It reads as a formulary or teaching aid, not as an account of original discoveries.
- **Oral transmission.** The mnemonic structure of f86v4 (Section 3), the anomalous hapax ratio, and the circular song format suggest a system designed to be sung, not merely read. This is consistent with an environment where written transmission was restricted or unreliable.
- **Deliberate concealment.** The encoding system (agglutinative cipher, non-representational botanical illustrations, chromatic therapeutic code) implies deliberate protection of the encoded knowledge. The concealment is structural, not decorative.

This functional profile—practical, compilatory, oral, concealed—is compatible with several historically attested contexts. The specific social and institutional identification of this context

is among the most consequential implications of the framework, but it is also the one most vulnerable to premature closure. It is therefore registered here as a research direction, not as a conclusion. The full chain of convergence (visual, structural, distributional) that supports this direction will be presented in dedicated future work.

15 Concluding Note: A Call to Resolution

This report maps the implications that the pharmaceutical framework generates, assigns each an epistemic status, and fixes their priority in a dated, falsifiable form.

The framework, if substantially correct, makes a set of questions *formulable for the first time*:

- What specific pharmaceutical tradition—in terms of period, region, and school—produced a system that organizes knowledge by solvent class, therapeutic class, and processing sequence in precisely this way?
- What are the social conditions under which a comprehensive pharmaceutical formulary would require the level of concealment documented in this manuscript?
- How does the gallows ductus system (H8b) relate to the physical layout of a medieval pharmaceutical workspace?
- What known historical texts from the *aqrābādhīn* genre show analogous structural signatures when analyzed with the same distributional methods?

These questions were not askable before the framework existed. They are now. Their resolution requires expertise that exceeds any single researcher's competence: medieval Arabic pharmacology, codicology, paleography, historical botany, and the history of cryptographic systems.

This report therefore constitutes an explicit invitation. The statistical engine is published in Paper I. The morphological framework and its implications are documented here. What remains—the full exposition of each convergence path, the detailed visual evidence, the broader interpretive consequences of the reading—will be presented through a structured program of dedicated studies, collaborative verification, and scholarly presentation. The author welcomes falsification attempts, independent replication, and any form of engagement that treats the framework as what it is: a testable hypothesis.

Appendices (Companion Repository)

The following appendices are maintained in the companion repository rather than in this document, to allow for continuous updating as the research progresses:

- **Appendix A. Morpheme Dictionary.** Inventory of 42+ morphemes with corpus counts, distributional evidence, and confidence tier for each entry.
- **Appendix B. Falsification Register.** Complete table: hypothesis, session of origin, evidence that prompted it, evidence that falsified it, session of falsification, replacement (if any), current status.
- **Appendix C. Plant Identification Dossiers.** One-page summary per plant: folio, textual processing profile, predicted pharmacological properties, sealed-envelope prediction, IIIF illustration comparison, convergence score.
- **Appendix D. High-Resolution IIIF Figures.** f86v4 four cardinal figures, f85r rosette grid, selected botanical folios with chromatic annotations, Cosmo bundle details.

- **Appendix E. Replication Guide.** How to reproduce every corpus-based result in this report.

All appendices are available at https://github.com/alessandroplaca-uro/voynich-pharmacists-cipher (Paper I scripts and data) and at the companion repository (to be released upon publication of this report).

Data Availability

The IVTFF interlinear transcription corpus is publicly available at https://www.voynich.nu/. Analysis scripts for Paper I are available at https://github.com/alessandroplaca-uro/voynich-pharmacists-cipher. High-resolution manuscript images are available through the Yale Beinecke Library IIIF manifest at https://collections.library.yale.edu/manifests/2002046. The morpheme dictionary and plant identification dossiers referenced in the appendices above will be released in the companion repository upon publication.

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References

[1] al-Kindī, Abū Yūsuf Ya'qūb ibn Isḥāq (9th c.). *Kitāb Kīmiyā' al-'Iṭr wa-al-Taṣ'īdāt* (Book of the Chemistry of Perfume and Distillations). Ed. K. Garbers (1948). Leipzig: Brockhaus.

[2] al-Rāzī, Abū Bakr Muḥammad ibn Zakariyyā (10th c.). *Kitāb al-Ḥāwī fī al-Ṭibb* (The Comprehensive Book on Medicine). Hyderabad: Osmania Oriental Publications, 1955–1970.

[3] Amancio, D.R., Altmann, E.G., Rybski, D., Oliveira, O.N., & Costa, L.F. (2013). Probing the statistical properties of unknown texts: application to the Voynich Manuscript. *PLoS ONE*, 8(7), e67310.

[4] Bax, S. (2014). A proposed partial decoding of the Voynich Manuscript. Unpublished manuscript, Bedfordshire, UK.

[5] Currier, P.H. (1976). New research on the Voynich Manuscript: Proceedings of a seminar. Communication presented at a seminar held on November 30, 1976.

[6] D'Imperio, M.E. (1978). *The Voynich Manuscript: An Elegant Enigma*. Fort George G. Meade, MD: National Security Agency.

[7] Ibn Sīnā (Avicenna). *Al-Qānūn fī al-Ṭibb* (The Canon of Medicine). Translated selections consulted via Bakhtiar, L. (1999). *The Canon of Medicine*. Chicago: Great Books of the Islamic World.

[8] Montemurro, M.A., & Zanette, D.H. (2013). Keywords and co-occurrence patterns in the Voynich Manuscript: an information-theoretic analysis. *PLoS ONE*, 8(6), e66344.

[9] Placa, A. (2026). The Pharmacist's Cipher: Six Statistical Tests Supporting a Pharmaceutical Reading of the Voynich Manuscript (MS 408). Preprint. DOI: 10.5281/zenodo.19197846.

[10] Sherwood, E., & Schwartz, R.K. (2014). Botany and the Voynich Manuscript. Unpublished manuscript.

[11] Stolfi, J. (2005). A quantitative study of the Voynich Manuscript. Technical report, Institute of Computing, University of Campinas.

[12] Tucker, A.O., & Talbert, R.H. (2013). A preliminary analysis of the botany, zoology, and mineralogy of the Voynich Manuscript. *HerbalGram*, 100, 70–85.

[13] Zandbergen, R. (2022). The Voynich Manuscript. https://www.voynich.nu/